

Neutron Strategic Technology
Advanced Research
(Neutron STAR)
Advanced Research Announcement (ARA)



Industry User Guide

14 Mar 2023

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***The Government reserves the right to amend this user guide as necessary. The most current version will be updated to Government wide Point of Entry (GPE).**

CHAPTER 1: INTRODUCTION

1. Overview: 10 U.S.C.§4023 – Procurement for Experimental Purposes “The Secretary of Defense and the Secretaries of the military departments may each buy ordnance, signal, chemical activity, transportation, energy, medical, space-flight, and aeronautical supplies, including parts and accessories, and designs thereof, that the Secretary of Defense or the Secretary concerned considers necessary for experimental or test purposes in the development of the best supplies that are needed for the national defense.”

2. Criteria for Use: Use of an ARA to solicit for research and development is encouraged when:

- There is an increased focus on mission integration or need for an agile contracting methods for more advanced R&D programs
- The technology is a covered area under the five (5) topic areas listed in the solicitation and 10 U.S.C.§4023
- Necessary for experimental or test purposes in the development of the best supplies and designs for national defense

CHAPTER 2: ARA SOLICITATION VARIATIONS: Variations of the ARA Solicitation process are available for use. The following types are commonly used, but are not considered all inclusive. (See Exhibit A for graphic depiction of some common ARA variations.)

Open Solicitation: This approach allows for white paper submittal at any time within the ARA active period. ARAs must be updated no less frequently than annually to remain active. White papers are no less than bi-annually.

Individual Solicitation: This approach allows for white paper and/or proposal submittal at a specified date and time as set forth in the ARA. Late bid and proposal provisions are usually included in the ARA.

One-Step: The one-step process is used to request full technical and cost proposals from each Offeror. The proposals are reviewed in accordance with the solicitation criteria and typically, all of a selected proposal, part of a selected proposal, or none of the proposals may be selected for award.

Two-Step: The two-step process is sometimes used when a large number of proposals are anticipated. Potential Offerors are invited to submit brief descriptive white papers in lieu of full proposals. The ARA must state whether an unfavorable white paper review will bar the Offeror from further consideration. Full proposals are requested from those Offerors selected in the white paper review process. When proposals are received, they are reviewed consistent with the process for one-step ARAs.

Limited Solicitation: This approach allows for a direct request of a Request for Proposal (RFP) or a white paper without publishing to the GPE.

The Government may issue solicitations that combine one or more of the above approaches.

CHAPTER 3: OPEN ARA WHITE PAPER PREPARATION

Offeror Submission Information

1. Overview: The submission process consists of two steps* described in detail below. White papers – and full proposals, if requested by the Government – shall be submitted in accordance with this announcement.

The Government anticipates white papers will be unclassified. **Offerors shall NOT submit classified white papers or proposals to the Neutron STAR Org Mailbox.** Special instructions will be provided in the event a classified proposal is required.

Dialogue between prospective Offerors and Government representatives is strongly encouraged. Any Offeror intending to submit a white paper in response to a topic area is encouraged to notify the Contracting Officer/Agreements Officer (CO/AO) in advance. This is a Two-Step Process; full proposals will not be accepted or evaluated unless specifically requested by the CO/AO.

Due to the dynamically evolving nature of US warfighter nuclear deterrent mission, the Neutron STAR announcement is subject to updates, revisions, and/or changes to the topic areas. Therefore, Offerors are highly encouraged to continually monitor <https://sam.gov> to ensure they receive latest guidance for this announcement.

2. White Paper (First Step) Submission Instructions:

a. **General:** This open announcement is currently requesting white papers and a rough order of magnitude (ROM) cost estimate in response to Topic 1, Topic 2, and Topic 5 of the five (5) topic areas outlined in the announcement. The white paper shall include a discussion of the nature and scope of the research and the Offeror's proposed technical approach. All white papers will be kept on file for one year. An Offeror submitting a proposal without first submitting a white paper **will not** be eligible for an award. The Government will not reimburse any Offeror the cost of preparing white papers in response to this announcement, nor is it considered an allowable direct charge to any resulting award or any other contract.

b. **Page Limitation:** The white paper shall be limited to **six (6)** full 8.5 inch x 11 inch pages, prepared and submitted in Microsoft Word or Adobe Acrobat. The font shall be Times New Roman 12 point with no less than one inch (1") margins (left, right, top and bottom). Headers and/or footers may be included in the margin parameters. Standard single line spacing shall be used for all paragraphs. All white paper pages shall be presented in portrait page layout orientation. Charts, graphs, matrices, tables, and similar depictions (including spreadsheets) shall be clearly legible. All electronic data must be virus free. The page limitation covers all information including indices, photographs, foldouts (counted as one (1) page for each 8.5 inch by 11 inch portion), tables, charts, appendices, attachments, etc. The Government will not consider pages in excess of these limitations.

WHITE PAPER SUBMISSION ORG BOX:

afrl.rvkb.afrlneutronstardms@us.af.mil

If it is determined that a classified white paper is required, please DO NOT deliver any classified portion of the white paper, but rather contact the Neutron STAR Org Mailbox for delivery instructions. If files contain links, the links must be intact and maintained through all revisions. Files shall not be read/write/password protected (i.e., must be unlocked, non-password protected, and/or unprotected).

c. **Format:** The white paper shall be formatted as follows:

(1) **Cover Sheet:** The cover sheet is not included in the page limitation and should be complete upon submission.

(2) **Section A. White Paper Narrative:** The narrative should include but not be limited to the following:

a.) **Abstract:** Identify the relevant topic areas applicable to the white paper and provide a brief (500 word maximum) summary of the overall proposed technical effort. The abstract should summarize or highlight key facets of the proposed effort. The Government intends to utilize the abstract to more efficiently route the white paper submission to correct evaluators/technical divisions for consideration.

b.) **Discussion of the nature and scope of the research; the problem statement; the Offeror's proposed objectives and technical approach/solution; all relevant assumptions; an overview of tasks and methods to achieve each objective; and expected results.**

c.) **Description of state-of-the-art knowledge or innovative methods to be employed, along with a clear strategy, resources, and schedule for maturing the technology. Address if this research has been currently, or previously, performed with another agency.**

d.) **Description of final product(s) to be delivered (prototypes, demos, reports, software, etc.)**

e.) **Mapping of effort to specific need(s) outlined in the topic area(s) description, with a strategy for transition of deliverables to a Government acquisition program. Transitions can be specific to an existing/future system or have broad application to several existing/future systems and must include a schedule estimate.**

f.) **Explanation of how the proposed effort provides benefits to the topic area(s) needs and a strategy for verifying those intended benefits.**

g.) **Any restrictions on Government use, release, or disclosure of technical data.**

h.) **Key personnel, consultants, or subcontractors to support the effort and evidence of expertise and experience to accomplish the proposed tasks.**

- (3) **Section B. White Paper ROM Cost:** Provide a ROM cost estimate for planning purposes. The ROM should provide top-level details on major cost items such as labor, special equipment that will need to be purchased, materials, and any subcontracts or other organizations' costs (to include government furnished facilities, equipment, or personnel) for the proposed effort. The ROM cost estimate will be used to assess best value to the Government (cost versus benefit) and determine if there are sufficient funds available for the proposed effort. Any second step RFPs will require a full cost proposal.

d. **Other Information:** If the Offeror wishes to restrict its white papers, it shall be clearly and appropriately marked with the restrictions. For FAR-based proposals, follow the guidance under FAR 52.215-1(e).

NOTE: White paper may not be read if all requirements of this announcement are not met.

NOTE: Offerors are advised that duplicate white paper submissions received within the same 12-month period will not receive separate evaluations. Please do NOT submit duplicate white paper submission or white papers that would be considered substantially the same effort within the same 12-month period.

3. Proposal (Second Step) Submission Instructions: Full proposal instructions will be provided within a Request for Proposal Letter from the CO/AO as applicable and will not be posted to the GPE.

*** The Government reserves the right to collapse a white paper/Two-Step process into a One-Step process if the situation warrants and the white paper contains sufficient technical detail to do full Technical Proposal Evaluation instead of a white paper Evaluation.**

CHAPTER 4: COMMUNICATION

All communications with the Government program manager or Acquisition Team members must follow communication guidelines as follows:

- Communication with Acquisition Team members is encouraged prior to the issuance of an RFP, after which time all communication must be submitted, in writing, to the Contracting Office.
- Acquisition Team members cannot attempt to replace the potential Offeror's original ideas with their own ideas or the program office's preferred approach.
- Acquisition Team members cannot share ideas or technical solutions provided in conversations, white papers, or proposals to a competing potential proposer or anyone else outside the team.
- If the Acquisition Team member provides information concerning the objectives/goals/requirements of the ARA to one potential Offeror, this information must be provided to all potential Offerors with the CO/AO's coordination via an ARA amendment.
- If a potential Offeror is provided with information that expands/contradicts/retracts information contained in the published announcement, this same information must be made publicly available to all potential Offerors in a Questions and Answers document. In some cases, this will also require an amendment to the ARA.
- The Acquisition Team members may discuss whether the potential Offeror is "on-track" and/or consistent with the needs of the AF. However, while the potential Offeror may be "on-track" is no way intended to imply the potential Offeror will be awarded a contract or issued an RFP.
- Any one-on-one communications are not a pre-screening process.
- The Acquisition Team members cannot answer requests for advice regarding technical approach / desired solution or individual proposal elements, e.g., subcontractors, materials, labor hours, etc.
- Notwithstanding anything discussed during a one-on-one communication, all terms and conditions of the announcement remain unchanged unless and until an amendment is issued.

NOTE: Additional Pre-proposal communication procedures maybe established at the Individual Solicitation level by the CO/AO.

CHAPTER 5: INDIVIDUAL SOLICIATION PROCESS

Individual Solicitations under this ARA will be announced on the GPE and, for assistance instruments, Grants.gov. Solicitations will request interested Offerors to submit either a white paper or full proposal (depending on the ARA Solicitation variation selected, see Chapter 2). The Individual Solicitation announcement on the GPE and, for assistance instruments, Grants.gov will be the only solicitation. This may be a short announcement that links to further information or the full announcement. It is recommended interested Offerors read each Individual Solicitation very carefully. Typically, the following information may be included in Solicitation announcements:

- An identifying Individual Solicitation number and program name.
- Points of contact for both contracting and technical matters. Offerors are encouraged to contact the listed points of contact for any assistance or questions.
- A description of the stated areas of potential research or a description of the scientific or engineering problems needing new and creative solutions. A short summary of areas of program interest, expanded as appropriate, to include problems, objectives, and deliverable items (reports, software, prototypes, etc.).
- Information on proposal or white paper format and submission requirements.
- **The address and/or method for submittal.**
- If a two-step variation is used, a description of the full proposal invitation procedures.
- **Criteria for selecting white papers and proposals.**
- If an informational briefing is scheduled, details as to time, date, place, number of attendees permitted, clearances needed, etc.
- Advice to foreign-owned firms that their participation is subject to foreign disclosure review procedures and that they should immediately contact the contracting focal point for information if they contemplate responding.
- If export-controlled technical data is involved, a note advising that only firms holding certification under the US/Canada Joint Certification Program (JCP) <http://www.dla.mil/HQ/InformationOperations/Offers/Products/LogisticsApplications/JCP.aspx> are allowed access to such data.
- The total dollar value or range of dollar values as well as anticipated period of performance.
- A statement that multiple white papers or proposals addressing different topic areas, within the purview of the announcement, may be submitted by each Offeror.
- Notice to Offerors that only Contracting/Grants/Agreements Officers are legally authorized to commit the Government.
- Date and time when white papers or full proposals will be due.

CHAPTER 6: REVIEW INFORMATION

1. Evaluation Criteria: Evaluations for white papers and proposals submitted will be conducted by a peer or scientific review process and will use the following criteria:

a. **Technical Merit:** Technical Merit is first in order of priority (more important than Relevance to AFRL Programs or Value/Cost) and shall be evaluated based on the following criteria, which are of equal importance.

- (1) Overall scientific or technical merit
 - Offeror demonstrates innovation
 - Offeror demonstrates knowledge and application of state-of-the-art methods and/or technologies
 - Problem statement, objectives and deliverables are clearly defined
 - Objectives and expected results are anchored by theory, analysis or data
 - Assumptions are logical, reasonable and appropriate
- (2) Relevance to mission needs outlines in topic area description
 - Proposed effort is logically mapped to one or more topic area needs
 - Proposed effort provides valuable contribution to meeting one or more topic area needs
 - Benefits appropriately off-set identified risks
 - Rough Order of Magnitude (ROM) schedule estimate meets mission needs
- (3) Soundness of the Offerors Technical Approach (For Proposals Only)
 - Approach is defined, logical and feasible
 - Schedule, speed of delivery (tech readiness date) and required resources are reasonable and appropriate
 - Proposed results are measurable and testable
 - Key staff have appropriate and qualifications to perform proposed effort

b. **Value/Cost:** Value/Cost is subject to funds availability.

- (1) **For white papers:** The ROM cost estimate will be evaluated on whether it is appropriate for the expected benefit and if sufficient funding is available. White papers that satisfy the technical merit factor, demonstrate benefits that are commensurate with the cost, and for which sufficient funding is determined to be available will be evaluated as best value to the Government.
- (2) **For proposals:** Cost will be evaluated on the degree to which it is realistic, reasonable and represents the best value to the Government. Cost realism will assess the likelihood that the proposed technical approach can be successfully accomplished for the cost proposed. Cost reasonableness will consider the cost proposed in conjunction with program objectives and availability of funding. Prior to award of a potential successful offer, the Contracting Officer will make a final determination regarding price reasonableness.

NOTE: Evaluation criteria for Individual Solicitations may vary. If so, specific evaluation criteria will

be addressed at the Solicitation level.

2. White Paper Review and Selection Process (First Step): Evaluation of white papers will be conducted by a peer or scientific review process, using the criteria described above (or as outlined at the Individual Solicitation level). Based on the evaluation, white papers will be categorized as Best Value, Valuable, Not Recommended. White papers that are evaluated as Best Value may be sent a Request for Proposal Letter from the CO/AO. The Government's goal is to notify Offerors of the disposition of their white paper within 3 months after White Paper submission at minimum. Unsuccessful Offerors will be notified in writing and will be provided with a consensus report of the evaluation. Valuable awards not finalized due to lack of funding will be retained for 12 months for consideration once funding is secured.

- a. **Best Value**: Scientifically and technically sound approach relevant to mission needs and funding available
- b. **Valuable**: Scientifically and technically sound approach relevant to mission needs but no funding available
- c. **Not Recommended**: Not a scientifically and technically sound approach and/or not relevant to mission needs

Under the Open ARA, White Papers may be submitted at any time; however, the submissions will be evaluated within 3 months after White Paper submission at minimum.

3. Proposal Review and Selection Process (Second Step): Proposals requested after white papers will also be reviewed against the criteria described above. Based on the peer or scientific review process, proposals will be categorized as:

- a. **Highly Recommended**: Well-conceived scientifically and technically sound proposal pertinent to accomplish goals and objectives. Proposals are recommended for acceptance (subject to availability of funds) and normally are displaced only by other Highly Recommended proposals.
- b. **Selectable**: Scientifically or technically sound proposals that may need additional development. Proposals are recommended for acceptance if sufficient funding is available, but at a lower priority than Highly Recommended proposals. To ensure a diversity of approaches, a Selectable proposal may be prioritized over a Highly Recommended proposal if the Selectable proposal presents a unique approach unlike any of the Highly Recommended proposals.
- c. **Not Selectable**: Proposal is not technically sound or does not meet agency needs. Even if sufficient funding existed, the proposal will not be funded. Not selectable proposals will not be retained.

Note 1: The recommendations above will be based on the Government's evaluation of the technical approach independent of the cost proposal. If recommended for award, the Contracting Officer will make a final determination regarding price reasonableness.

Note 2: When the Government elects to award only part of a proposal, the selected part may be categorized as Highly Recommended or Selectable although the proposal as a whole may not merit such a categorization.

Note 3: The Government may award some, all, or none of the proposals received, and may award without

negotiation/discussion; however, the Government reserves the right to negotiate with any Offeror(s) whose proposal is selected for funding.

CHAPTER 7: AWARD PROCESS (for Open White Paper Solicitation and Individual Solicitations)

1. Notification of Award Decision: The CO/AO or CS will notify, in writing, any Offeror(s) selected for negotiation of potential award and any Offeror(s) that was not recommended for award. Such notification will only occur after the technical peer or scientific review has been approved. The notification of recommendation for award is not to be construed to mean an award is assured, as availability of funds and successful negotiations are prerequisites to any award. During this notification process, the CO/AO/CS may request any required updates or revisions begin immediately.

a. **More Than One Award and Flexibility in Awards made:** Under this ARA, more than one award may be made based on the quality of the proposals and availability of funding. Occasionally, the Government may be interested in awarding only a certain portion (or portions) of a white paper or proposal. This ARA method provides the flexibility to make an award for only those portions or tasks of the white paper or proposal that are of interest to the Government. The Offeror will be notified in writing if the Government intends to make an award based on the white paper or proposal. The notification will indicate if all or only portions of the proposal will be included in the award.

b. **Sequence of Awards:** Awards will be made to Highly Recommended/Selectable proposals. However, because of the uniqueness of the ARA review process, it may be the case that while the overall proposal is ranked Not Selectable a certain part of it may be ranked Highly Recommended or Selectable. In this case the flexibility of the ARA process will allow the Government to make an award for that portion of the proposal that is of high interest to the Government. Not Selectable proposals, with the exception of certain parts of the proposal (as described above) are generally not awarded. Offerors whose proposal(s) are not recommended for acceptance will be notified by the CO/AO.

2. Model Document: The CO/AO/CS may transmit the Government's anticipated award format (Model Document) to the selected Offeror(s). The model document provides the basis for negotiations on all award requirements, terms, and conditions, and maybe attached to the Individual Solicitation announcement or RFP.

3. Technical and Cost Negotiations

a. **Technical:** Any unresolved technical issues will normally be addressed first. Typically, negotiations involve specific language issues or determining Government needs versus effort proposed. Most of these negotiations are conducted via telephone/online meeting resources and supplemented by email. If many complex issues arise, the Government negotiator may request face-to-face meetings to resolve them.

b. **Terms and Conditions:** Any exceptions, changes, or additions to the model document should be clearly identified and justification should be provided. Some may affect pricing or technical considerations (e.g., limited and restricted rights, licensing agreements, or royalties) and should normally be resolved before negotiating cost/fee/profit and total price.

c. **Cost/Fee/Profit (if applicable):** Price should be addressed last. If rates, factors, or other cost

elements change after proposal submission, the CO/AO must be informed as quickly as possible to preclude any delays. The Offeror may be asked to document these changes. The CO/AO is responsible for evaluating the reasonableness of offered prices. The analytical techniques and procedures employed are meant to ensure that the final price is fair and reasonable.

d. **Agreement:** Once negotiations have been concluded, the CO/AO will ask the Offeror to confirm the negotiations in writing. The confirmation should include the cost/fee/profit or price, and any terms and conditions agreed upon. Depending on the award type (e.g. FAR based, OT, Grant, etc.) the CO/AO will finalize any pre-award requirements and submit to the Offeror for final execution.

4. Science & Technology (S&T) Program Protection - For Proposal Responses Only

Offerors that are submitting proposals shall submit as a part of the proposal submission 1.) a completed Standard Form (SF 424), Research and Related Senior/Key Person Profile (Expanded) Form (See Attachments) for all Senior/Key Personnel proposed; and, 2.) A Completed Security Program Questionnaire (See Attachments). Offerors may be asked to provide a mitigation plan for any identified S&T Protection risks. The Government reserves the right to determine an offeror unawardable on the grounds of unacceptable S&T Protection risk based on its review of the SF 424 and Security Program Questionnaire.

By submitting a proposal, the offeror certifies that it is in compliance with Section 223(a) of the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021 which requires that: (a) the Principal Investigator (PI) and other key personnel certify that the current and pending support provided on the proposal is current, accurate and complete; (B) agree to update such disclosure at the request of the agency prior to the award of support and at any subsequent time the agency determines appropriate during the term of the award; and (c) the PI and other key personnel have been made aware of the requirements under Section 223(a)(1) of this Act. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. code, Title 218, Section 1001).

This is not required for white paper submissions.

CHAPTER 8: ASSISTANCE INSTRUMENTS

1. Background: An award may be made from this open ARA announcement or any Solicitations in the form of a Grant, Cooperative Agreement (CA), or Other Transactions (OT) for Research, these are collectively known as "Assistance Instruments." Assistance Instruments are appropriate when the goal of the project is to provide assistance by transferring a thing of value (such as money or government facilities) to accomplish a public purpose. On the other hand, a FAR type contract is usually more appropriate when the goal of the program is the purchase of goods or services for the direct benefit of the government. Assistance Instruments, while legally termed contracts, are not subject to the FAR. There are three (3) basic kinds of Assistance Instruments: (1) Grants, (2) CAs, and (3) OTs for Research Projects (10 U.S.C. 4021 only).

2. General: Assistance Instruments differ from FAR type contracts in that they are usually more streamlined and less complex. Assistance Instruments establish a "partnership" between the Government and the recipient rather than the "buyer/seller" relationship common with a FAR contract. A grant does not require substantial Government involvement, whereas a CA or OT for research does. "Substantial Government involvement" is defined as involvement, of a technical nature, over and above the normal program management functions. Assistance Instruments are similar to FAR contracts in that they also require basic cost and technical proposals (including the requirement for a proposed Statement of Work [SOW]), as well as negotiation and agreement on articles, similar to terms and conditions. They differ in that agreement is based on an estimated budget rather than a fair and reasonable price. They are covered by OMB guidance, "Uniform Administrative Requirements, Cost Principles, and Audit Requirements for Federal Awards," published in the CFR at 2 CFR part 200 and implemented by the DoD at 2 CFR part 1103, "Interim Grants and Cooperative Agreements Implementation of Guidance in 2 CFR part 200" (79 FR 76047, December 19, 2014). Provisions of the DoDGARs other than 32 CFR parts 32 and 33 continue to be in effect, with applicability as stated in those provisions and are not subject to the FAR.

3. When a Grant Would Be Awarded: Grants are normally awarded for basic and applied research efforts. They are used when the objective is to advance the state of knowledge in an area of science or technology; to find a solution to a specific problem; or to achieve improved performance or reduce cost where no specific end result or product, other than documentation, is required. Any hardware involved would be for demonstration or proof-of-principle purposes only. The award of a Grant must meet several other requirements in addition to the above. A Grant must provide to the Government, at a minimum, a paid-up license and march-in rights in patents. The following characteristics apply to a Grant award:

- a. Primarily awarded to educational or non-profit organizations.
- b. Classified information is usually not involved.
- c. There is no fee or profit.
- d. 2 CFR 200 (2 CFR 1103) and DoDGARs (other than parts 32 and 33) applies.
- e. Substantial involvement between the Government and recipient is not required.
- f. Not subject to the terms of a procurement contract.

4. When a CA Would be Awarded: CAs are normally awarded for basic research, applied research, or advanced research or development efforts. The following characteristics apply to a

CA award:

- a. Used when a FAR type contract or grant has been determined inappropriate.
- b. Substantial involvement between the Government and recipient is required.
- c. Paid-up license and march-in rights in patents must be obtained.
- d. Government Purpose Rights in data usually obtained.
- e. There is no profit/fee.
- f. Cost share may be highly encouraged or required and may consist of other than cash; IR&D cost sharing allowable.
- g. 2 CFR 200 (2 CFR 1103) and DoDGARs (other than parts 32 and 33) applies.

CHAPTER 9: OTHER TRANSACTIONS & MISCELLANEOUS

1. Background: This announcement, along with Individual Solicitations, allows for the award of Other Transactions under the authorities of 10 U.S.C. §4021-Research Projects and 10 U.S.C. §4022 - Prototype Project. FAR requirements and processes are not applicable to awards issued under these authorities. As such, all articles, terms and conditions will be negotiated on a case-by-case basis with the CO/AO.

2. 10 U.S.C. §4021 - Research Projects: Awards under this authority can be considered either an assistance instrument (see Chapter 8 above) where the principal purpose is stimulating a public purpose or as an acquisition where the main objective is a direct benefit of the Government. Regardless of the type, awards under this authority shall be for basic, applied and advanced research projects, and generally have a focus on dual-use (commercial and defense) applications.

3. 10 U.S.C. §4022 - Prototype Projects: Offerors that are contemplating a response under this authority shall refer to the latest DoD Other Transaction Guide for additional information and guidance.

- a. The statute allows for prototype projects “that are directly relevant to enhancing the mission effectiveness of military personnel and the supporting platforms, systems, components, or materials proposed to be acquired or developed by the Department of Defense, or to improvement of platforms, systems, components, or materials in use by the armed forces.”
- b. Prototypes Projects address: “proof of concepts, models, reverse engineering to address obsolesce, pilots, novel application of commercial technologies for defense purposes, agile development activity, creation, design, development, demonstration of technical or operational utility, or combinations of the foregoing. A process, including a business process, may be the subject of a prototype project.”
- c. To be eligible for award, transactions under this authority must meet one of the requirements as identified under 10 U.S.C. §4022(d)(1), summarized here:
 - (1) At least one nontraditional defense contractor or nonprofit research institution participating to a significant extent
 - (2) All significant participants in the transaction are small businesses or nontraditional contractors
 - (3) At least one third of the total cost of the prototype is to be paid out of funds provided by sources other than the Federal Government
 - (4) The senior procurement executive determines in writing that exceptional circumstances justify use of a transaction that would not be feasible or appropriate under a contract
- d. In accordance with 10 U.S.C. §4022 (f), a prototype project awarded under this authority may provide for the non-competitive award of a follow-on production contract or transaction to the participants in the transaction. Non-competitive procedures are allowed if (summarized):
 - (1) Competitive procedures were used for the selection of parties in the transaction; and

(2) The participants in the transaction successfully completed the prototype project
Offerors are hereby notified that the Government is definitively interested in the award of a follow-on transaction or award for all awarded prototype projects unless explicitly stated otherwise within an Individual Solicitation.

- e. Successful completion criteria for the prototype project includes (or as otherwise defined within an Individual Solicitation):
 - (1) Met key technical goals of a project,
 - (2) Satisfied success metrics incorporated into the Prototype award, or
 - (3) Accomplished a particularly favorable or unexpected result that justifies transition to production or follow-on activities

CHAPTER 10: DEFINITIONS

10 U.S.C. §4023 – Procurement for Experimental Purposes

“The Secretary of Defense and the Secretaries of the military Departments may each buy ordnance, signal, chemical activity, transportation, energy, medical, space-flight, and aeronautical supplies, including parts and accessories, and designs thereof, that the Secretary of Defense or the Secretary concerned considers necessary for experimental or test purposes in the development of the best supplies that are needed for the national defense.”

Advisory and Assistance Services (A&AS): Those services provided under contract by nongovernmental sources to support or improve: Organizational policy development; decision-making; management and administration; program and/or project management and administration; or R&D activities. It can also mean the furnishing of professional advice or assistance rendered to improve the effectiveness of Federal management processes or procedures (including those of an engineering and technical nature). In rendering the foregoing services, outputs may take the form of information, advice, opinions, alternatives, analyses, evaluations, recommendations, training and the day-to-day aid of support personnel needed for the successful performance of ongoing Federal operations.

Assistance Instrument: Business arrangement that is appropriate when the goal of the program is to transfer a thing of value to accomplish a public purpose of support or stimulation as authorized by law. Assistance instruments include grants, cooperative agreements, technology investment agreements, and other transactions for research. Assistance instruments are covered by Office of Management and Budget (OMB) guidance, "Uniform Administrative Requirements, Cost Principles, and Audit Requirements for Federal Awards," published in the Code of Federal Regulations (CFR) at 2 CFR part 200 and implemented by the DoD at 2 CFR part 1103, "Interim Grants and Cooperative Agreements Implementation of Guidance in 2 CFR part 200" (79 FR 76047, December 19, 2014). Provisions of the DoD Grants and Agreements Regulations (DoDGARs) other than 32 CFR parts 32 and 33 continue to be in effect, with applicability as stated in those provisions and are not subject to the Federal Acquisition Regulation

Contract Specialist (CS): Term used interchangeably to include the Contract Negotiator or Buyer. Be advised only a Contracting Officer is warranted to obligate the Government. Regulations specify several other functions that must be performed by a Contracting Officer.

Contracting/Agreements Officer (CO/AO): Title given to a person with the authority to enter into, administer, and/or terminate contracts or agreements and make related determinations and findings. Only a Contracting/Agreements Officer has the authority to obligate the Government.

Cooperative Agreement: An assistance instrument in which there will be substantial involvement between the Government and the recipient to perform basic research, applied research, and advanced technology development. A cost-sharing arrangement may be required. (See 2 CFR §200.24)

Cost Realism Analysis: The process of independently reviewing and evaluating specific elements of each Offeror's proposed cost estimate to determine whether the estimated proposed cost elements are realistic for the work to be performed; reflect a clear understanding of the requirements; and are consistent with the unique methods of performance and materials described in the Offeror's technical proposal. Cost realism analysis shall be performed on cost-reimbursement contracts to determine the probable cost of performance for each Offeror. (See FAR 15.404-1(d).)

Government wide Point of Entry (GPE): The single point where Government business opportunities greater than \$25,000, including synopses of proposed contract actions, solicitations, and associated information, can be accessed electronically by the public. The GPE is (See FAR 2.101). The GPE is located at <https://www.sam.gov>

Government Furnished Property (GFP): Property in the possession of, or directly acquired by, the Government and subsequently furnished to the contractor for performance of a contract. Government-furnished property includes, but is not limited to, spares and property furnished for repair, maintenance, overhaul, or modification. Government-furnished property also includes contractor-acquired property if the contractor-acquired property is a deliverable under a cost contract when accepted by the Government for continued use under the contract. (See FAR Part 45.)

Grant: An instrument under which the Government provides assistance to a recipient for basic, applied or advanced research. There is no substantial Government involvement. (See 2 CFR §200.51)

Grants.gov: The public notification media used when an ARA will allow for the award of an assistance instrument (grant, cooperative agreement, technology investment agreement).

Independent Research and Development (IR&D) Cost: The cost of effort which is neither sponsored by a grant, nor required in performing a contract, and which falls within any of the following four areas: (a) Basic research, (b) Applied research, (c) Development, and (d) Systems and other concept formulation studies. (See FAR 31.001.)

Other Transactions for Prototype 10 U.S.C. §4022 provides the Secretary of the Air Force authority to enter into transactions (other than contracts) for prototype projects that are directly relevant to enhancing the mission effectiveness of military personnel and the supporting platforms, systems, components, or materials proposed to be acquired or developed by the Department of Defense, or to improvement of platforms, systems, components, or materials in use by the armed forces. Agreements issued under this authority are commonly referred to as Other Transaction (OT) for Prototype agreements. (See *The OSD Other Transaction Guide for Prototype Projects* for more information)

Other Transactions for Research: An instrument under which the Government provides assistance to a recipient (can be a single contractor or a team of contractors) to transfer something of value to the public sector. Substantial Government involvement and 50/50 cost sharing are required. Used when it is not appropriate or feasible to use a contract, grant, or

cooperative agreement. (See 10 U.S.C. §4023)

Peer or Scientific Review: The process where each proposal is reviewed against the criteria called out in the ARA. It may also address the technical team's assessment of applicable quantitative elements, to include number of labor hours proposed, skills mix, materials, subcontracting, etc.

Price Analyst: A contracting professional whose function is focused on analyzing an Offeror's cost proposal and documenting the results of the negotiation. A Price Analyst is not a Contract Specialist.

Program Manager (PM): The individual responsible for overseeing/managing a Science & Technology (S&T) acquisition or assistance project. This individual may have a different job title in their primary position.

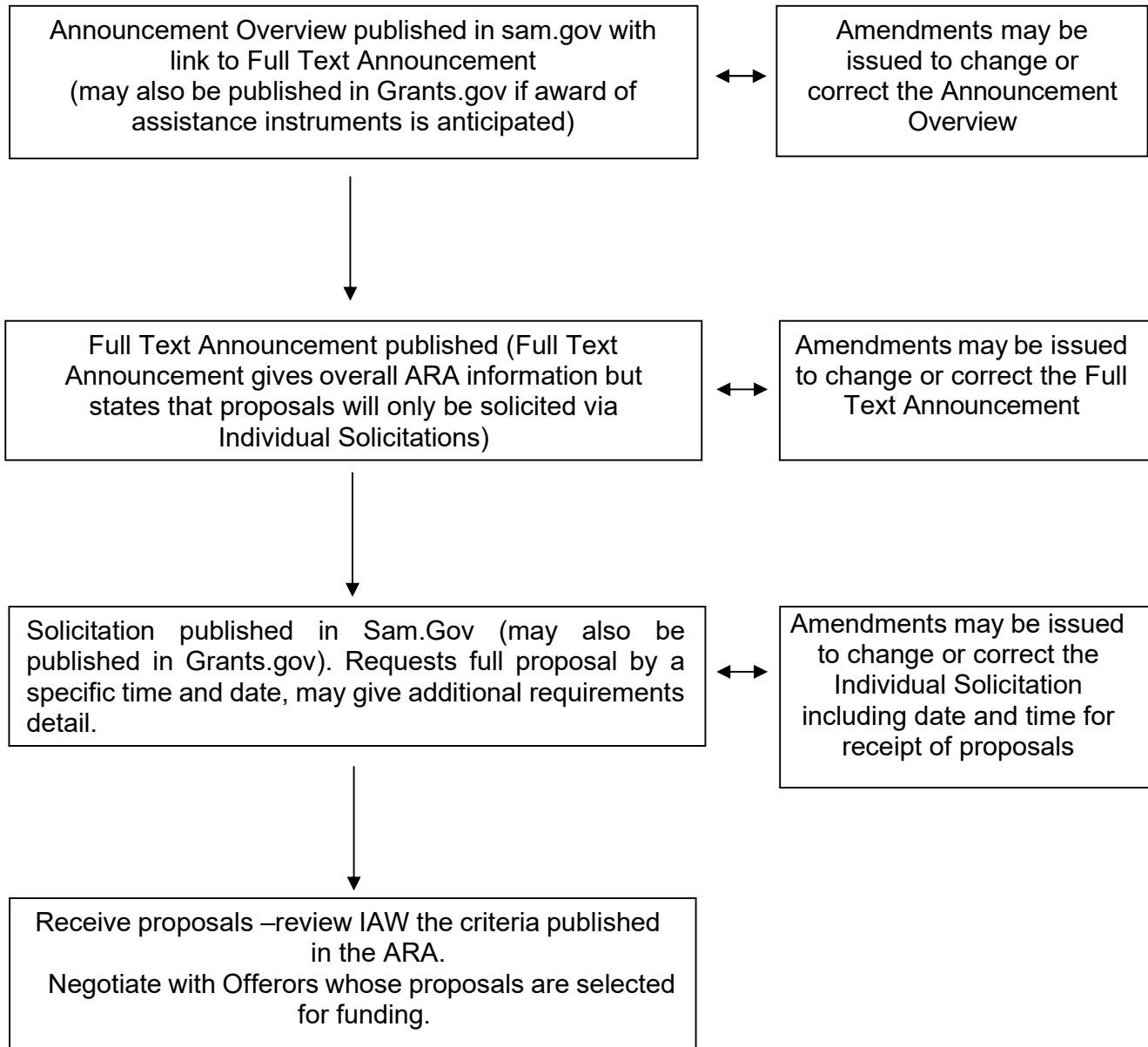
Statement of Work (SOW): The document that captures the technical content of the contractual effort. The contract may incorporate portions of the Offeror's technical proposal instead of a formal SOW. For ARAs, Offerors are often required to submit a non-proprietary contractor statement of work (sometimes referred to as a C-SOW) when responding to the RFP.

Technical/Cost Peer or Scientific Review: The team's review of each proposal against each criterion. The team is required to document this review and provide a recommendation to the Contracting Officer.

White Paper: A brief summary of the proposed technical approach with an accompanying rough-order-of-magnitude (ROM) price.

EXHIBIT A: GRAPHIC DEPICTIONS OF COMMON ARA SOLICIATION VARIATIONS

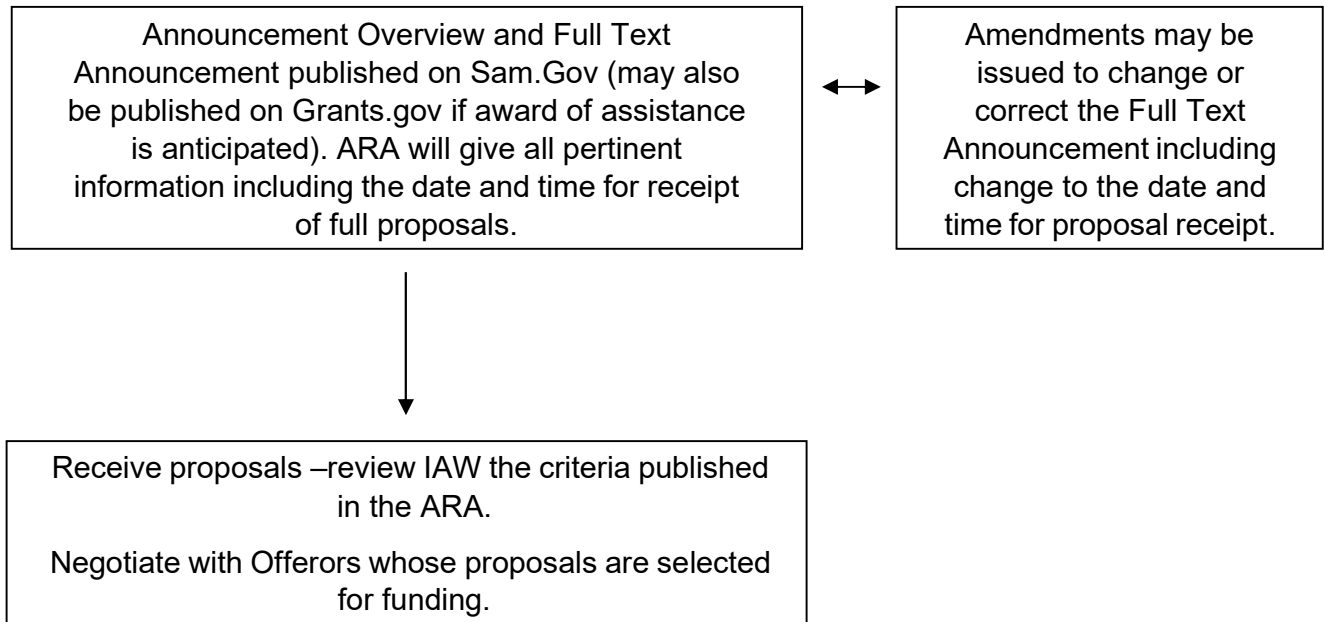
ARA with Solicitations (Neutron STAR)



Two-Step Open Solicitation



One-Step Open Solicitation



ARA Open White Paper Solicitation

